MAGMAFUNCTIONS MAGMA package documentation

Lucien François

Version: 02nd July 2026

Introduction and contents

This document provides documentation for the functions in the package "MAGMAFUNCTIONS.m" that contains functions related to linear algebra, combinatorics and finite fields to be used on MAGMA. This package was build by Lucien FRANÇOIS during its doctoral program in the University College Dublin. The functions have been tested on MAGMA version `Magma V2.28-16`.

The references relevant for the mathematical aspects of the functions below are given in the end of the document.

Documentation in progress, sorry for the disturbance.

1 Arithmetics

MF_PrimePowersUpTo(m):

Given an integer m, returns the sequence of prime powers at most m, i.e. returns the sequence of all elements in $\{p^\eta : p \text{ prime}, \eta \in \mathbb{N}, p^\eta \leq m\}$. The asymptotic complexity of the function is the one of the MAGMA function `PrimesUpTo`.

Input	**Output**
• `m` : an integer.	A sequence of integers (`SeqEnum` of `RngIntElt`).

```
> MF_PrimePowersUpTo(100);
[ 2, 3, 4, 5, 7, 8, 9, 11, 13, 16, 17, 19, 23, 25, 27, 29, 31, 32, 37, 41, 43, 47, 49, 53, 59,
61, 64, 67, 71, 73, 79, 81, 83, 89, 97 ]
```

2 Combinatorics

2.1 Bijections

`MF_IntToqary(n, q, size):`

Given n a non-negative integer, q an integer with $q \geq 2$, and *size* a positive integer, returns the image of n through the bijection $[\![0, q^{size} - 1]\!] \xrightarrow{\sim} [\![0, q-1]\!]^{size}$ that writes the integer n into the basis q. That is the output is $(a_i)_{i \in [\![0, size-1]\!]}$ such that $n = a_0 + a_1 q + \cdots + a_{size-1} q^{size-1}$.

Input	Output
• `n` : a non-negative integer, • `q` : an integer at least 2, • `size` : a positive integer.	A sequence of integers (`SeqEnum` of `RngIntElt`) of size `size`.

```
> MF_IntToqary(50,2,10);
[ 0, 1, 0, 0, 1, 1, 0, 0, 0, 0 ]
```

`MF_qarytoInt(seq, q):`

Given *seq* a sequence of non-negative integers and q an integer with $q \geq 2$ and strictly larger than every entry of *seq*, returns the image of *seq* through the bijection $[\![0, q-1]\!]^{size} \xrightarrow{\sim} [\![0, q^{size} - 1]\!]$ such that $(a_i)_{i \in [\![0, size]\!]} \mapsto a_0 + a_1 q + \cdots + a_{size-1} q^{size-1}$, where *size* is the length of the sequence *seq*.

Input	Output
• `seq` : A sequence of integers (`SeqEnum`), • `q` : an integer at least $\max(2, \max($`seq`$))$.	An integer (`RngIntElt`).

```
> MF_qarytoInt([ 0, 1, 0, 0, 1, 1, 0, 0, 0, 0 ],2);
50
```

`MF_IntToFq(n, q):`

....

Input	Output
• ...	An integer (`RngIntElt`).

```
```

`MF_FqtoInt(x):`

....

Input	Output
• ...	An integer (`RngIntElt`).

`MF_IntToVSpace(n, q, Vdim):`

....

Input	Output
• ...	An integer (`RngIntElt`).

`MF_VSpaceToInt(x):`

....

Input	Output
• ...	An integer (`RngIntElt`).

`MF_IntToIntTuple(N,i):`

`MF_IntTupleToInt(N,i):`

2.2 Counting in algebra

`MF_GBC(q, n, r):`

Given a prime power q and two non-negative integers n and r, returns the Gaußian binomial coefficient

$$\begin{bmatrix} n \\ r \end{bmatrix}_q := \begin{cases} \prod_{t=1}^{r} \frac{q^{n-t}-1}{q^t-1} & \text{if } r \leq n \\ 0 & \text{else.} \end{cases}$$

That is returns the number of r-dimensional subspaces of the vector-space $\mathbb{F}_q^n$.

Input	Output
• `q` : a prime power, • `n` : a non-negative integer, • `r` : a non-negative integer.	An integer (`RngIntElt`).

```
> MF_GBC(8,5,3);
304265
```

MF_NbMatrixRank(n, m, q, r):

Given non-negative integers n, m, and r with a prime power q the number of matrices in $\mathbb{F}_q^{n\times m}$ of rank exactly r, that is to say the integer

$$\begin{bmatrix} m \\ r \end{bmatrix}_q \sum_{k=0}^{r} (-1)^{r-k} \begin{bmatrix} r \\ k \end{bmatrix}_q q^{nk+\binom{r-k}{2}} = \begin{bmatrix} m \\ r \end{bmatrix}_q \prod_{i=0}^{r-1} (q^n - q^i).$$

This functions needs the function MF_GBC. See [VLW92, Theorem 25.2].

Input	Output
• n : a non-negative integer, • m : a non-negative integer, • q : a prime power, • r : a non-negative integer.	An integer (RngIntElt).

```
> MF_NbMatrixRank(6,5,3,2);
639518880
```

MF_NbTrankOne(q,N):

Given a prime power q and $N = [N_1, \ldots, N_m]$ a non-empty sequence of positive integers, returns the number of tensors of size $N_1 \times \cdots \times N_m$ over the field $\mathbb{F}_q$ of rank 1, that is to say the integer $(q-1)^{m-1} \prod_{j=1}^{m} (q^{N_j} - 1)$. This functions needs the function MF_GBC. See [VLW92, Theorem 25.2].

Input	Output
• q : a prime power, • N : a non-empty sequence of positive integers.	An integer (RngIntElt).

```
> MF_NbTrankOne(2,[2,3,4,5]);
9765
```

MF_NbTrankTwo(q,N):

Given a prime power q and $N = [N_1, \ldots, N_m]$ a non-empty sequence of positive integers **of length at most** 3, returns the number of tensors of size $N_1 \times \cdots \times N_m$ over the field $\mathbb{F}_q$ of rank exactly 2. if $m = 1$, it returns 0. If $m = 2$, it returns MF_NbMatrixRank(N_1,N_2,q,2). For $m = 3$, it returns the value computed in [BCF26, Lemma 5.1].

Input	Output
• q : a prime power, • N : a sequence of positive integers of length 1, 2 or 3.	An integer (RngIntElt).

```
> MF_NbTrankTwo(2,[2,3,4]);
32970
```

MF_MRDrankdistribution(q,m,n,d,r) :

Given a prime power q, given m, n, r non-negative integers and d a positive integer, returns the number of matrices of rank exactly r in an $[m\times n, \max(m,n)(\min(m,n)-d+1), d]_q$ maximum rank-distance rank-metric code, that is to say a subspace $\mathcal{C} \leq \mathbb{F}_q^{m\times n}$ of minimum rank d and dimension $\max(m,n)(\min(m,n)-d+1)$. This functions needs the function `MF_GBC`. See [Gor21, Theorem 11.4.15].

Input	**Output**
• `q` : a prime power, • `m` : a non-negative integer, • `n` : a non-negative integer, • `d` : a postive integer, • `r` : a non-negative integer.	An integer (`RngIntElt`).

```
> for r in [0..5] do  printf "\%o, ", MF_MRDrankdistribution(2,6,5,3,r); end for;
1, 0, 0, 9765, 97650, 154728,
```

References

[BCF26] Eimear Byrne, Alain Couvreur, and Lucien François. Decoding algorithms for tensor codes. *arXiv:2604.16105*, 2026.

[Gor21] Elisa Gorla. Rank-metric codes. In *Concise Encyclopedia of Coding Theory*. Chapman and Hall/CRC Press, 2021.

[VLW92] J. H. Van Lint and R. M. Wilson. *A Course in Combinatorics*. Cambridge University Press, Cambridge, 1992.